Universidad de Dagupan

Software Engineering 1

Software Requirements Specification Document

Daló

Daló is a QR-code-based event registration and attendance system built for university student organizations. It replaces the traditional pen-and-paper sign-in process—and its long onsite queues—by letting students register for an event online beforehand and receive a unique QR code, which event facilitators scan on the day of the event to confirm attendance instantly and accurately.

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31-CSE-01

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1. Introduction

Purpose of the Document

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of Daló, a QR-code-based event registration and attendance system for university student organizations. It exists to give a single, unambiguous reference for what the system must do before implementation begins, so that design decisions, database structure, and code can be traced back to a real, stakeholder-verified need rather than assumption. This document is intended for the course instructor evaluating the project, and for the developer (myself) as the authoritative reference throughout design, implementation, and testing.

Scope of the Document

Daló replaces the manual, pen-and-paper (or screenshot-based) event registration and attendance process currently used by student organizations, which causes long onsite queues and unreliable, disputable attendance records. The system allows a student to register for an event online and receive a unique QR code, and allows an event facilitator to scan that code at the door to confirm attendance in real time. The value delivered is a faster, more accurate check-in process for organizers and a queue-free experience for students. The system will be developed as a solo, zero-budget student project using entirely free tools (PHP, MySQL, free-tier hosting, and free student-licensed software), over the course of one academic term, from initial specification through a working deployed demo.

2. General Description

Daló's general function is to digitize two linked processes—event registration and attendance verification—using a QR code as the bridge between them. The primary objective for each user group differs: a student wants to register quickly and prove attendance without waiting in line; an event organizer wants an accurate, real-time count of who actually attended versus who merely registered; a facilitator wants to confirm attendance in seconds per person, even under poor signal or low phone battery.

The intended user community is university students and student organization officers—a technically comfortable but not necessarily technical audience, all of whom already own smartphones with browsers and cameras. No app installation is required; every interaction happens through a mobile web browser. The core benefit over the current manual process is the elimination of file-upload friction (no screenshots to submit) and the elimination of disputed attendance (a QR code can only be scanned once, removing the manual-error source both interviewees independently identified).

3. Stakeholder Table

Role	What They Need	How They Were Reached
Student / Attendee	A fast, low-friction way to register for events and prove attendance without queueing or uploading screenshots/proof of registration.	Interview—Hermione Chan, student, conducted 08/31/2026.
Event Organizer / Officer	An accurate, real-time way to see who has actually shown up versus who merely registered, and to eliminate disputes caused by manual verification errors.	Interview—Christian Joseph Cabubas, Phinma CITE Student Council, conducted 08/31/2026.
Facilitator / Door Staff	A quick scan-and-confirm method that works even under poor signal or low phone battery, so the check-in line keeps moving.	Derived from both interviews (officer's onsite-slowdown account and student's connectivity concern); role confirmed by proponent based on lab requirements.

4. Functional Requirements

ID	Requirement
FR-01	The system shall allow an event organizer to create an event with a title, date, time, location, and registration capacity.
FR-02	The system shall allow a student to register for an event through an online form without uploading any photo, screenshot, or file as proof.
FR-03	The system shall generate a unique QR code for each student immediately upon successful registration.
FR-04	The system shall allow a registered student to view and download their QR code.
FR-05	The system shall allow a facilitator to scan a student's QR code using a phone's browser camera to confirm attendance.
FR-06	The system shall record attendance in real time within seconds of a successful QR scan.
FR-07	The system shall reject a QR code that has already been scanned and shall notify the facilitator that the student is already checked in.
FR-08	The system shall display a live dashboard to the organizer showing the number of attendees present versus the total number registered.
FR-09	The system shall automatically close registration once the event's set capacity is reached.
FR-10	The system shall allow a facilitator to manually search for and check in a student by name when a QR code cannot be scanned.
FR-11	The system shall send the student a confirmation containing their QR code once registration is complete.

ID	Requirement
FR-12	The system shall allow the organizer to export the final attendance list as a report after the event ends.

5. Interface Requirements

User Interfaces

Daló is browser-based and mobile-first. Four primary screens are required: (1) an event registration form for students, (2) a QR code display/download page shown after successful registration, (3) a camera-based scanning page for facilitators, and (4) a live dashboard for organizers showing registered-vs-present counts.

Software Interfaces

The PHP application layer communicates with MySQL via mysqli or PDO, using prepared statements for all queries. QR code generation is handled server-side in PHP using the phpqrcode library (a single-file, dependency-free library well-suited to shared hosting without Composer/SSH access). QR code scanning is handled client-side in the browser using the html5-qrcode JavaScript library, which accesses the device camera via the browser's getUserMedia API, decodes the QR code locally, and sends the decoded value to a PHP endpoint via fetch/AJAX as JSON for validation against the database.

Communication Interfaces

All communication occurs over HTTPS between the student/facilitator's mobile browser and the web server. The application will be reachable at a dedicated subdomain (e.g., dalo.ljdevera.dev), pointed via DNS to the hosting provider.

6. Performance Requirements

Static Requirements

The database must be able to store records for at least 500 attendees per event without redesign, and the system must support multiple concurrent events without data collision between them.

Dynamic Requirements

A QR scan must be confirmed and recorded within 3 seconds for 95% of scans; the registration form must load and accept submission within 2 seconds under normal conditions; the system must correctly handle at least 200 concurrent users without a measurable increase in response time; the error rate for duplicate or rejected check-ins must be 0% for already-scanned codes (this must always be caught, not merely minimized).

7. Design Constraints

- **Algorithm/Library Constraints:** QR generation is constrained to phpqrcode (no Composer/build step, compatible with free shared hosting); QR scanning is constrained to a browser-camera-based JavaScript library (html5-qrcode) rather than a native app, since no app store distribution is in scope.
- **Hardware/software limitations:** the system must run on standard free-tier shared PHP hosting (typically PHP 8.x, MySQL 5.7+/8.x, no SSH access, no Composer). It must function on any modern smartphone browser with camera permission granted—no dedicated scanner hardware is assumed.
- **Standards and policy:** handling of student personal data (name, student ID) must conform to Republic Act No. 10173, the Data Privacy Act of 2012.
- **Security requirements:** all database queries must use parameterized/prepared statements to prevent SQL injection; each QR code must encode a long, non-sequential, unique token (not a guessable ID like an auto-increment number) to prevent forgery or enumeration of other students' codes.
- **Resource limits:** the project has a $0 budget, constraining hosting to free-tier providers, which typically cap storage, bandwidth, and database size—the schema and image assets must be kept lightweight accordingly.
- **Operating environment:** deployment target is a subdomain of an existing personally-owned domain, DNS-routed separately from the domain's existing GitHub Pages portfolio site.

8. Preliminary Schedule and Budget

Schedule

- **October (Midterm):** SRS finalized; ER diagram completed (entities, PK/FK, cardinalities).
- **November (Semifinal):** Layered architecture diagram completed; core implementation begins—registration, QR generation, database schema live on free hosting.
- **December–January (Finals):** QR scanning/attendance flow, live dashboard, manual fallback check-in completed; source code finalized with README; live/recorded demo prepared.

Budget

$0 total. All tools used are free: PHP, MySQL, and Composer-free libraries (phpqrcode, html5-qrcode) cost nothing; the domain was obtained free via the GitHub Student Developer Pack; hosting will use a free-tier PHP/MySQL provider; development tools (VS Code, GitHub, phpMyAdmin) are all free or already covered under existing student licensing.

9. Non-Functional Requirements

ID	Quality Attribute	Requirement (with measure)
NFR-01	Performance	A QR scan shall be confirmed and recorded within 3 seconds, for 95% of scans, to prevent the check-in queue from building up.
NFR-02	Usability	A student shall be able to complete event registration in under 2 minutes without submitting any file or screenshot.
NFR-03	Reliability	The check-in process shall remain functional under intermittent internet connectivity through a manual name-search fallback, addressing low-signal or low-battery scenarios.
NFR-04	Security	Each QR code shall be uniquely generated per registrant and shall not be reusable for a second check-in, preventing duplicate or disputed attendance records.
NFR-05	Scalability	The system shall support check-in of at least 500 attendees within a 30-minute window without a drop in scan response time.
NFR-06	Compliance	Collection and storage of student personal data (name, ID number) shall conform to Republic Act No. 10173, the Data Privacy Act of 2012.

10. User Stories with Acceptance Criteria

US-01 [Must]

As a student, I want to register for an event without uploading any photo or screenshot, so that I don't struggle with slow uploads like I did before.

Acceptance criteria: GIVEN an event is open for registration, WHEN the student submits the registration form, THEN a QR code is generated and made available without requiring any file upload.

US-02 [Must]

As a student, I want to receive a unique QR code right after registering, so that I can present it at the door instead of standing in a long verification line.

Acceptance criteria: GIVEN a completed registration, WHEN the registration is submitted successfully, THEN a unique QR code is generated and delivered to the student.

US-03 **[Must]**

As a facilitator, I want to scan a student's QR code using my phone's browser camera, so that I can confirm attendance in seconds instead of manually checking screenshots.

Acceptance criteria: GIVEN a valid, unscanned QR code, WHEN the facilitator scans it, THEN attendance is marked and confirmed within 3 seconds.

US-04 **[Must]**

As a facilitator, I want the system to reject an already-scanned QR code, so that duplicate check-ins and attendance disputes don't happen.

Acceptance criteria: GIVEN a QR code already marked present, WHEN it is scanned again, THEN the system shows 'already checked in' and does not create a duplicate record.

US-05 **[Should]**

As an event organizer, I want a real-time dashboard of attendees present versus registered, so that I know the exact turnout during the event.

Acceptance criteria: GIVEN the event is ongoing, WHEN a student is checked in, THEN the dashboard count updates immediately.

US-06 **[Should]**

As a facilitator, I want to manually search and check in a student by name when their QR code won't load, so that a dead phone battery or no signal doesn't block their entry.

Acceptance criteria: GIVEN a student cannot display their QR code, WHEN the facilitator searches their name, THEN the facilitator can mark them present manually.

US-07 **[Should]**

As an event organizer, I want to set a maximum capacity that closes registration automatically, so that I don't over-enroll the event.

Acceptance criteria: GIVEN a capacity limit is set, WHEN registrations reach that limit, THEN the registration form closes and displays 'Registration full.'

US-08 **[Could]**

As an event organizer, I want to export the attendance list after the event, so that I can submit reports without manually recounting names.

Acceptance criteria: GIVEN the event has ended, WHEN the organizer requests an export, THEN a report of all checked-in attendees is generated.

Why US-01/US-02/US-03 are the top Must-haves: both interviewees independently identified the same two failures—slow onsite proof-checking (organizer) and friction-heavy file uploads during registration (student). A QR code that is generated without file uploads and scanned in seconds directly removes both root causes; without this core loop working, the application has no reason to exist over paper.

11. Process Model Recommendation

Recommended model: Agile (Scrum, short sprints).

Scope stability—Daló is intentionally generic so it can serve many different student organizations across the university, and each organization may surface different edge cases (e.g., approval workflows, ticketed events) once they start using it. This evolving scope favors Agile over Waterfall's frozen-requirements approach.

Cost of a wrong release—this is not a safety-critical or regulated system; a missed edge case costs a support fix, not a compliance incident. This removes the main justification for Waterfall's heavy up-front gates.

Customer availability—both stakeholder types (a student council officer and a general student) are classmates and org-mates who are reachable weekly for feedback on each increment, which is exactly what Agile assumes and Waterfall does not require.

Pipeline maturity—as a single-semester student project without an established CI/CD pipeline, full DevOps is not yet achievable; Agile/iterative delivery (working increments each sprint: registration → QR generation → scanning → dashboard) is the realistic fit.

Appendix: Elicitation Notes

Interview 1 — Event Organizer

Interviewee: Christian Joseph Cabubas, Phinma CITE Student Council

Date: August 31, 2026 **Technique:** Interview

Q: Walk me through how you currently register people for an event, start to finish.

A: The event is announced first, then the important details and registration information are sent out to the target participants. On the event day, registration is confirmed onsite by checking screenshots or other proof of registration, which can sometimes be unreliable.

Q: What breaks or slows down on event day?

A: The actual confirmation of registrations on-site usually slows down the process, especially when there are many attendees.

Q: How do you currently check who actually showed up vs who just registered?

A: Usually through another online form that attendees have to answer to confirm their attendance.

Q: What information do you need to see instantly during the event (capacity left, who's arrived)?

A: The number of attendees who are actually present compared to the total number of people who registered.

Q: Have you ever had someone dispute their attendance was recorded?

A: Yes. There have been multiple cases, usually because of errors during the attendance process.

Q: What would make you trust a digital system over paper?

A: If the system provides accurate information and real-time feedback, especially when confirming registrations and attendance.

Interview 2 — Student Attendee

Interviewee: Hermione Chan, Student

Date: August 31, 2026 **Technique:** Interview

Q: Describe your worst experience registering/queueing for a campus event.

A: It was probably because of an early cut-off, and then people kept cutting in line.

Q: How long did you wait in line, and why?

A: For campus events, it probably takes about 2 to 3 hours. To be honest, I'm not really sure why the line moves so slowly, or maybe they were short-staffed so it took longer to accommodate everyone.

Q: Would you be comfortable showing a QR code on your phone at the door? What if you have no signal or low battery?

A: Yes, I am comfortable.

Q: What would make you actually complete a pre-registration instead of skipping it?

A: It's convenient and pretty straightforward, but sometimes it takes longer because you need to upload different pictures, and my Google Drive wasn't cooperating.

Q: Have you ever been denied entry due to a registration mix-up?

A: I haven't experienced that yet.